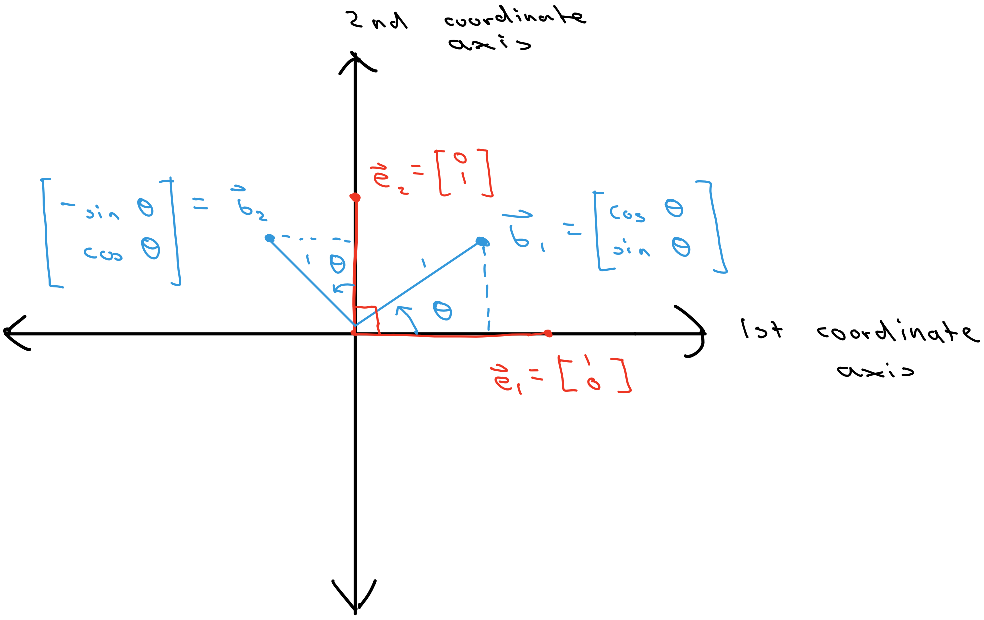
Derivation of Givens Rotations



Let's construct a matrix

$Q \in \mathbb{R}^{2\times 2}$ so that

i. $[Q]_{2\times 2}[\vec{e}_1]_{2\times 1} = [\vec{b}_1]_{2\times 1}$

ii. $[Q]_{2\times 2}[\vec{e}_2]_{2\times 1} = [\vec{b}_2]_{2\times 1}$

Let's set $Q = \begin{bmatrix} q_{11} & q_{12} \\ q_{21} & q_{22} \end{bmatrix}_{2\times 2}$

Claim.

We can derive our Givens Rotation Matrix by using the linear combination of column vectors definition of matrix-vector multiplication.

Consider

i.

$$Q \cdot \vec{e}_1 = \left[\begin{array}{c|c} q_{11} & q_{12} \\ q_{21} & q_{22} \end{array}\right] \begin{bmatrix} 1 \\ 0 \end{bmatrix} = 1 \cdot \begin{bmatrix} q_{11} \\ q_{21} \end{bmatrix} + 0 \cdot \begin{bmatrix} q_{12} \\ q_{22} \end{bmatrix} = \begin{bmatrix} q_{11} \\ q_{21} \end{bmatrix} = \begin{bmatrix} \cos\theta \\ \sin\theta \end{bmatrix}$$

ii.

$$Q \cdot \vec{e}_2 = \left[\begin{array}{c|c} q_{11} & q_{12} \\ q_{21} & q_{22} \end{array}\right] \begin{bmatrix} 0 \\ 1 \end{bmatrix} = 0 \cdot \begin{bmatrix} q_{11} \\ q_{21} \end{bmatrix} + 1 \cdot \begin{bmatrix} q_{12} \\ q_{22} \end{bmatrix} = \begin{bmatrix} q_{12} \\ q_{22} \end{bmatrix} = \begin{bmatrix} -\sin\theta \\ \cos\theta \end{bmatrix}$$

$$\Rightarrow \quad Q_\theta = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

This is a Givens rotation by angle θ in the counterclockwise direction.

$\Rightarrow$ clockwise rotation by θ (is counterclockwise by $-\theta$)

$$\Rightarrow \quad Q_{-\theta} = \begin{bmatrix} \cos -\theta & -\sin -\theta \\ \sin -\theta & \cos -\theta \end{bmatrix} = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}$$

$$= Q^T_\theta$$

Q^T and Q are known as orthogonal matrices. (unitary matrices)

$$I_n = Q^T Q$$

$$= Q Q^T$$